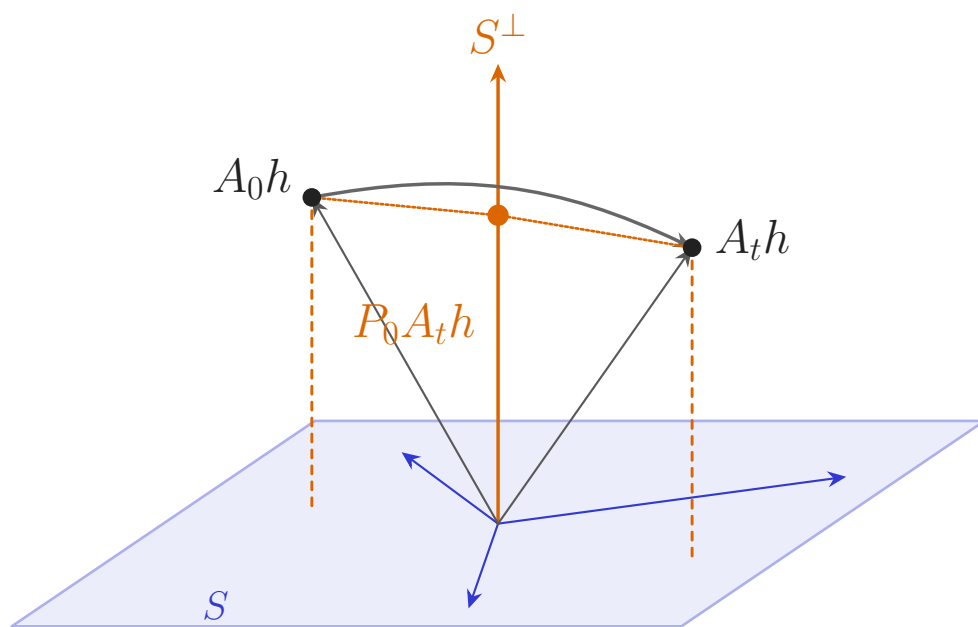


The preserved subspace and the certificate

During SGD

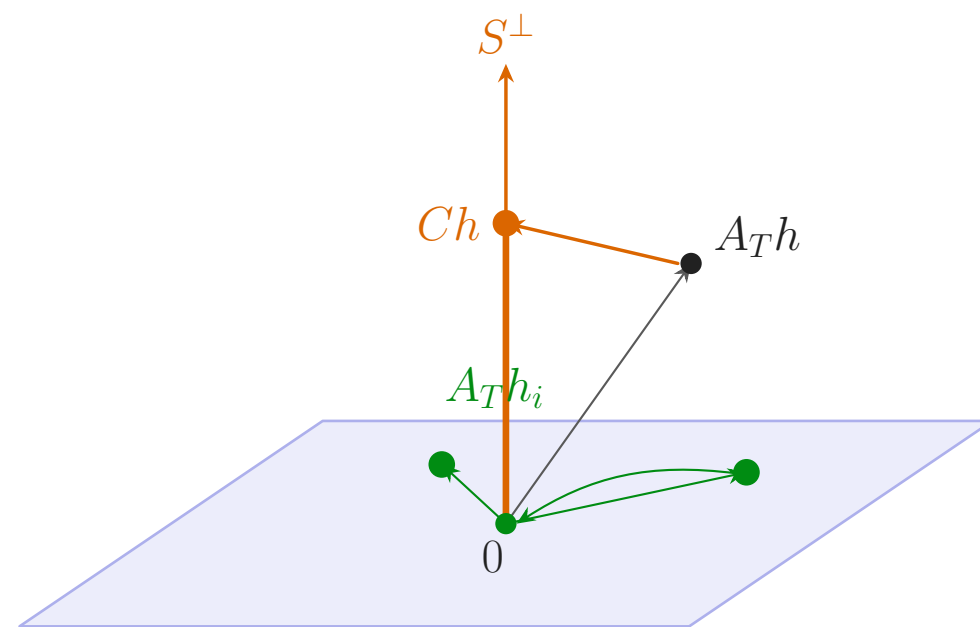


$$S = \text{col}(A_0H)$$

$$B_t P_0 = 0$$

$$P_0 A_t = P_0 A_0$$

At release



$$S = \text{row}(B_T), \quad \dim S = q$$

$$C = P_0 A_T, \quad P_0 = P_{S^\perp}$$

$$C \phi(x_i) = 0, \quad \text{rank } C = r - q$$

$$H = [\phi(x_1) \cdots \phi(x_N)] \text{ fixed}, \quad \text{rank } B_T = q = \text{rank } H < r \leq n$$

plain SGD; $B_0 = 0$; generic independent A_0 ; no weight decay